

Elements of Statistics and Probability

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Table of Contents

1 Measures of Dispersion

- Measures of Dispersion for Raw Data
- Measures of Dispersion for Grouped Data

2 Skewness and Kurtosis

Motivation

- Measures of central tendency help us to represent the entire mass of the data by a single value
- Can the **central tendency** describe the data fully and adequately?
- Let us consider an example:

- The daily income (in £) of the workers in two factories are:

Factory A: 35 45 50 65 70 90 100

Factory B: 60 65 65 65 65 65 70

- Here, we observe that in both factories the mean income is the same (£65)

Motivation (cont.)

- In factory A, the observations are much more scattered from the mean
- In factory B, almost all the observations are concentrated around the mean
- The two groups differ even though they have the same mean
- Therefore, we need some other measures which concern with the measure of spread
- To do this, we study what is known as **measures of dispersion**

Measures of Dispersion

- It deals with the spread of the data
- A small value of the measure of dispersion indicates that the data are clustered closely
- A large values indicates knowing the estimate of the central tendency is not sufficient

Measures of Dispersion (cont.)

- Absolute measures:
 - **Range**
 - **Mean Deviation**
 - **Variance**
 - **Standard Deviation**
- Relative measures:
 - **Coefficient of Variation**

Range

- Range indicates the difference between the **highest** and **lowest** value of the dataset

$$\text{Range} = \text{Highest value} - \text{Lowest value}$$

- Example: 93, 103, 105, 110, 104, 112, 105, 90

$$\text{Range} = 112 - 90 = 22$$

- Properties:
 - It depends only on two values
 - It can be influenced by extreme values

Mean Deviation from Mean

- **Mean Deviation:** The **arithmetic mean** of the absolute values of the deviations of the observations from the arithmetic mean.

$$MD = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$$

- Properties:
 - MD uses all the values in the sample
 - Absolute values are difficult to work with

Mean Deviation from Mean (cont.)

- Example: 93, 103, 105, 110, 104, 112, 105, 90

$$\bar{X} = \Sigma X / n = 822 / 8 = 102.75$$

X	$X - \bar{X}$	$ X - \bar{X} $
93	-9.75	9.75
103	0.25	0.25
105	2.25	2.25
110	7.25	7.25
104	1.25	1.25
112	9.25	9.25
105	2.25	2.25
90	-12.75	12.75
		$\Sigma X - \bar{X} = 45$

- $$MD = \frac{\Sigma |X - \bar{X}|}{n} = \frac{45}{8} = 5.62$$

Variance

- Variance is the arithmetic mean of the squared deviations of observations from the mean
- **Population variance:** $\sigma^2 = \frac{\sum(X - \mu)^2}{N}$
- **Sample variance:** $S^2 = \frac{\sum(X - \bar{X})^2}{n - 1}$
- σ^2 is a **Parameter** and S^2 is a **Statistic**

Variance (cont.)

- Example: 93, 103, 105, 110, 104, 112, 105, 90.

$$\bar{X} = \Sigma X / n = 822 / 8 = 102.75$$

X	$X - \bar{X}$	$(X - \bar{X})^2$
93	-9.75	95.06
103	0.25	0.06
105	2.25	5.06
110	7.25	52.56
104	1.25	1.56
112	9.25	985.56
105	2.25	5.06
90	-12.75	162.56

$$\Sigma(X - \bar{X})^2 = 407.48$$

- Population Variance = $\Sigma(X - \mu)^2 / N = 407.48 / 8 = 50.94$
- Sample Variance = $\Sigma(X - \bar{X})^2 / (n - 1) = 407.48 / 7 = 58.21$

Variance (cont.)

- Working formula for population variance is:

$$\sigma^2 = \frac{\sum X^2 - N\bar{X}^2}{N}$$

- Working formula for sample variance is:

$$S^2 = \frac{\sum X^2 - n\bar{X}^2}{n - 1}$$

Variance (cont.)

- Example: 93, 103, 105, 110, 104, 112, 105, 90

$$\bar{X} = \Sigma X / n = 822 / 8 = 102.75$$

X	X^2
93	8649
103	10609
105	11025
110	12100
104	10816
112	12544
105	11025
90	8100
$\Sigma X = 822$	$\Sigma X^2 = 84868$

- $\sigma^2 = (\Sigma X^2 - N\bar{X}^2) / N = (84868 - 8 \times 102.75^2) / 8 = 50.94$

- $S^2 = (\Sigma X^2 - n\bar{X}^2) / (n - 1) = (84868 - 7 \times 102.75^2) / 7 = 58.21$

Standard Deviation

- Standard Deviation (SD) is the positive square root of the variance

- Population SD: $\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum(X - \mu)^2}{N}} = \sqrt{\frac{\sum X^2 - N\bar{X}^2}{N}}$

- Sample SD: $S = \sqrt{S^2} = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}} = \sqrt{\frac{\sum X^2 - n\bar{X}^2}{n - 1}}$

- σ is a **parameter** and S is a **statistic**

Standard Deviation (cont.)

- Example: 93, 103, 105, 110, 104, 112, 105, 90
- Population SD: $\sigma = \sqrt{\sigma^2} = \sqrt{50.94} = 7.14$
- Sample SD: $S = \sqrt{S^2} = \sqrt{58.21} = 7.63$

Measures of Dispersion for Grouped Data

- **Range:** The difference between the upper limit of the highest class and the lower limit of the lowest class
- **Mean Deviation:**

$$MD = \frac{\sum_i f_i |X_i - \bar{X}|}{\sum_i f_i},$$

where f_i the class frequency X_i is the class midpoint, and $\bar{X} = \Sigma fX / \Sigma f$ is the arithmetic mean

Measures of Dispersion for Grouped Data (cont.)

- **Variance:**

$$S^2 = \frac{\sum f(X - \bar{X})^2}{n - 1} = \frac{\sum fX^2 - n\bar{X}^2}{n - 1},$$

where f_i the class frequency X_i is the class midpoint, and $\bar{X} = \sum fX / \sum f$ is the arithmetic mean

- **Standard Deviation:** $S = \sqrt{S^2}$

Exercise

- A sample of 50 antique dealers in Bath city revealed the following sales (in 000 £) last year:

Sales	Number of dealers
100–120	5
120–140	7
140–160	9
160–180	16
180–200	10
200–220	3

- (a) Find **range**, **mean deviation**, **variance** and **standard deviation**

Exercise (cont.)

- Solution:**

Sales	Midpoint (X_i)	Number of dealers (f_i)	$f_i X_i$	$f_i X_i^2$
100–120	110	5	550	60500
120–140	130	7	910	118300
140–160	150	9	1350	202500
160–180	170	16	2720	462400
180–200	190	10	1900	361000
200–220	210	3	630	132300
		$\Sigma f_i = 50$	$\Sigma f_i X_i = 8060$	$\Sigma f_i X_i^2 = 1337000$

- $\bar{X} = \Sigma fX / \Sigma f = 8060 / 50 = 161.2$

- Range** = $220 - 100 = 120$

- Variance:**

$$S^2 = \frac{\Sigma fX^2 - n\bar{X}^2}{n - 1} = \frac{1337000 - (50 \times 161.2^2)}{50 - 1} = 769.96$$

- Standard deviation:** $S = \sqrt{S^2} = \sqrt{769.96} = 27.75$

Exercise (cont.)

Sales	Midpoint (X_i)	Number of dealers (f_i)	$ X_i - \bar{X} $	$f_i X_i - \bar{X} $
100–120	110	5	51.2	256
120–140	130	7	31.2	218.4
140–160	150	9	11.2	100.8
160–180	170	16	8.8	140.8
180–200	190	10	28.8	288
200–220	210	3	48.8	146.4
$\Sigma f_i = 50$			$\Sigma f_i X_i - \bar{X} = 1150.4$	

• **Mean deviation:** $MD = \frac{\sum_i f_i|X_i - \bar{X}|}{\sum f_i} = \frac{1150.4}{50} = 23.01$

Relative Dispersion

- The usual measures of dispersion cannot be used to compare the dispersion if the units are different, even if the units are same but means are different
- It reports variation relative to the mean
- It is useful for comparing distributions with different units

Coefficient of Variation

- Coefficient of Variation (CV): The CV is the ratio of the standard deviation to the arithmetic mean, expressed as a percentage:

$$CV = \frac{S}{\bar{X}} \times 100$$

- The CV shows the extent of variability of data in a sample in relation to the mean of the population
- It can be interpreted as the variation per unit mean of data

Coefficient of Variation (cont.)

- Example: The variation in the monthly income of **executives** is to be compared with the variation in incomes of **unskilled** employees. For a sample of 50 executives, $\bar{X} = \text{BDT } 50,000$ and $S = 5,000$. For a sample of 50 unskilled employees, $\bar{X} = \text{BDT } 2,200$ and $S = 220$
- Solution:
 - For executives,

$$CV = (S/\bar{X}) \times 100 = (5000/50000) \times 100 = 10\%$$

- For unskilled employees,

$$CV = (S/\bar{X}) \times 100 = (220/2200) \times 100 = 10\%$$

Exercise 1

- Two sections of 10 students each in semester 4 at BRAC University were given a common test in statistics (maximum marks 40). The scores of the students are given below:

Section 15: 6 9 11 13 15 21 23 28 29 35

Section 16: 15 16 16 17 18 19 20 21 23 25

- (a) Calculate a measure of **central tendency** for each section. Explain whether this measure is enough to describe the data.
- (b) Calculate the **range**, **mean deviation**, **variance**, and **standard deviation** for each section.
- (c) Compare the two sections using an appropriate measure.

Exercise 2

- In a study on effectiveness of a medicine over a group of patients, the following results were obtained:

Percentage of relief:	0–20	20–40	40–60	60–80	80–100
No. of patients:	10	10	25	15	40

- (a) Calculate the **range**, **mean absolute deviation**, **variance**, and **standard deviation**.

Exercise 3

- In a study on ages of mothers at the first child birth in a village, the following data were available:

Age (in years):	18–20	20–22	22–24	24–26	26–28	28–30	30–32
No. of mothers:	130	110	80	74	50	40	16

- (a) Calculate the **range**, **mean absolute deviation from median**, **variance**, and **standard deviation**.

Exercise 4

- The weekly sales of post card and postage stamp in a local post office were recorded as follow:

Post card:	50	75	27	63	27	28	56
Postage stamp:	150	200	125	310	330	250	225

- (a) Calculate the **range**, **mean deviation**, **variance**, and **standard deviation** for post card and postage stamp separately.
- (b) Find out which of the two products shows greater fluctuation in sales.

Table of Contents

1 Measures of Dispersion

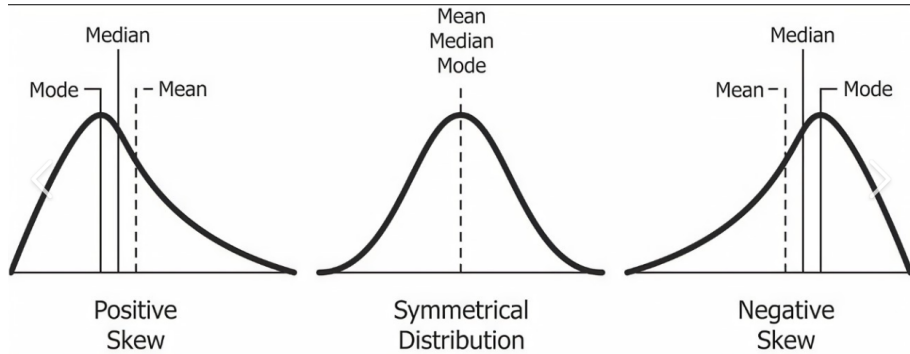
- Measures of Dispersion for Raw Data
- Measures of Dispersion for Grouped Data

2 Skewness and Kurtosis

Skewness

- Skewness is a measure of the asymmetry of a distribution
- A distribution can have right (or positive), left (or negative), or zero skewness (symmetric)
- A right-skewed distribution is longer on the right side of its peak, and a left-skewed distribution is longer on the left side of its peak

Skewness (cont.)



Skewness (cont.)

- For a positively skewed data, $\text{Mean} > \text{Median} > \text{Mode}$
- For a negatively skewed data, $\text{Mean} < \text{Median} < \text{Mode}$
- For a zero skewed data, $\text{Mean} = \text{Median} = \text{Mode}$

Sample Skewness

- Sample skewness:

$$S_k = \frac{1}{n} \times \frac{\sum_{i=1}^n (X_i - \bar{X})^3}{S^3}$$

- Here, n is the sample size, $\bar{X}$ is the sample mean and S is the sample standard deviation
 - If $S_k > 0.5$, then the distribution is positively skewed
 - If $-0.5 \leq S_k \leq 0.5$, then the distribution is approximately symmetrical
 - If $S_k < -0.5$, the distribution is negatively skewed

Other Measures of Skewness

- Pearson's first skewness coefficient (mode skewness):

$$\frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} = \frac{\bar{X} - M_0}{S}$$

- Pearson's second skewness coefficient (median skewness):

$$\frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}} = \frac{3(\bar{X} - M_e)}{S}$$

- If the measure is 0 (zero), then the data is zero skewed
- If it is > 0 (positive), then the data is positively skewed
- If it is < 0 (negative), then the data is negatively skewed

Sample Skewness (for Grouped Data)

- Sample skewness:

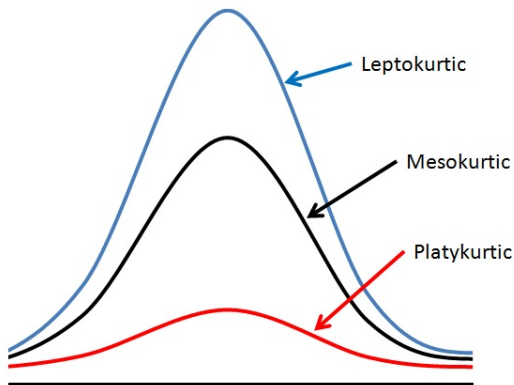
$$S_k = \frac{1}{n} \times \frac{\sum_{i=1}^n f_i (X_i - \bar{X})^3}{S^3}$$

- Here, n is the sample size, f_i is the **frequency**, X_i is the **class midpoint**, $\bar{X}$ is the sample mean and S is the sample standard deviation
 - If $S_k > 0.5$, then the distribution is positively skewed
 - If $-0.5 \leq S_k \leq 0.5$, then the distribution is approximately symmetrical
 - If $S_k < -0.5$, the distribution is negatively skewed

Kurtosis

- Kurtosis refers to measuring the degree to which a given distribution is more or less 'peaked' relative to the normal distribution
- In this regard, we have 3 categories of distributions:
 - Leptokurtic
 - Mesokurtic
 - Platykurtic

Kurtosis (cont.)



Sample Kurtosis

- Sample kurtosis:

$$S_{kr} = \frac{1}{n} \times \frac{\sum_{i=1}^n (X_i - \bar{X})^4}{S^4}$$

- Here, n is the sample size, $\bar{X}$ is the sample mean and S is the sample standard deviation
- The numerator and the denominator are always positive. Thus S_{kr} is always positive and is always measured relative to the kurtosis of a normal distribution, which is 3

Excess Kurtosis

- Excess kurtosis = $S_{kr} - 3$
 - If the excess kurtosis is > 0 (positive), then leptokurtic
 - If the excess kurtosis is 0 (zero), then mesokurtic
 - If the excess kurtosis is < 0 (negative), then platykurtic

Sample Kurtosis (for Grouped Data)

- Sample kurtosis:

$$S_{kr} = \frac{1}{n} \times \frac{\sum_{i=1}^n f_i (X_i - \bar{X})^4}{S^4}$$

- Here, n is the sample size, f_i is the **frequency**, X_i is the **class midpoint**, $\bar{X}$ is the sample mean and S is the sample standard deviation

Exercise 1

- Two sections of 10 students each in semester 4 at BRAC University were given a common test in statistics (maximum marks 40). The scores of the students are given below:

Section 15: 6 9 11 13 15 21 23 28 29 35

Section 16: 15 16 16 17 18 19 20 21 23 25

- (a) Calculate the sample skewness for each section and comment on its shape.
- (b) Calculate the sample kurtosis and compare the sections in terms of its peak.

Exercise 2

- In a study on effectiveness of a medicine over a group of patients, the following results were obtained:

Percentage of relief:	0–20	20–40	40–60	60–80	80–100
No. of patients:	10	10	25	15	40

- (a) Calculate Pearson's first and second skewness coefficients and comment on the shape of the data.
- (b) Calculate the sample skewness and sample kurtosis and comment on the shape of the data based on these two measures.

Thank You